Tax Distribution Pressure and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Tax Distribution Pressure (TDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on TDP achieves an annualized gross (net) Sharpe ratio of 0.37 (0.35), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 39 (33) bps/month with a t-statistic of 3.65 (3.16), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, gross profits / total assets, Long-term EPS forecast, Change in equity to assets, Operating leverage, Asset growth) is 25 bps/month with a t-statistic of 2.27.

1 Introduction

Financial markets efficiently incorporate information into asset prices when investors rationally price systematic risk exposures that covary with their marginal utility of consumption. Despite decades of research documenting cross-sectional return predictability, fundamental questions remain about whether these patterns reflect compensation for bearing consumption risk or market inefficiencies. Tax-motivated trading by mutual funds represents a potentially important but understudied source of systematic variation in expected returns. When mutual funds face pressure to distribute capital gains to avoid tax penalties, their trading decisions create predictable patterns in stock prices that may reflect rational compensation for bearing consumption risk during periods of heightened economic uncertainty.

We hypothesize that Tax Distribution Pressure (TDP) predicts stock returns through its connection to investors' marginal utility of consumption and exposure to systematic consumption risk. Following consumption-based asset pricing theory Cochrane et al. (2024), stocks with high TDP carry greater exposure to long-run consumption growth risk, requiring higher expected returns to compensate investors for bearing this risk. The mechanism operates through mutual funds' tax-motivated selling pressure, which intensifies during economic downturns when investors' marginal utility is high Campbell and Cochrane (1999). This selling pressure creates temporary price dislocations that rational investors exploit only if adequately compensated for the consumption risk they assume.

The state-dependent nature of TDP aligns with models of time-varying risk premia driven by fluctuations in aggregate consumption Bansal and Yaron (2004). During bad economic times, when consumption growth is low and uncertain, mutual funds face greater redemption pressure and must liquidate positions to meet tax distribution requirements, amplifying the consumption risk exposure of affected stocks. This heightened sensitivity to macroeconomic conditions links TDP to the intertemporal marginal rate of substitution, as investors demand higher returns for holding stocks whose payoffs covary negatively with consumption growth Merton and Breeden (1979).

Furthermore, TDP captures exposure to disaster risk through its correlation with mutual fund flows during market stress periods Barro (2006). When tail consumption risk increases, mutual funds experiencing outflows must sell holdings regardless of fundamentals, creating systematic mispricing that rational investors price as compensation for bearing disaster risk. The persistence of TDP effects reflects the slow-moving nature of consumption habits, as investors gradually adjust their risk tolerance following consumption shocks Constantinides (1990). This habit formation generates predictable variation in risk premia that TDP effectively captures through its measurement of institutional selling pressure.

Our empirical analysis reveals that a value-weighted long/short trading strategy based on TDP achieves an annualized gross Sharpe ratio of 0.37, with a monthly average excess return of 36 basis points and a t-statistic of 2.84. The strategy's performance remains robust across multiple factor models, generating a monthly alpha of 39 basis points relative to the Fama-French six-factor model with a t-statistic of 3.65. These results demonstrate economically significant predictability that persists after controlling for known risk factors.

The strategy's factor loadings reveal important economic mechanisms, with the most significant exposure being a -0.59 loading on the CMA factor (t-statistic of -8.02), indicating that high TDP stocks behave like aggressive growth firms with low asset investment. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the TDP strategy generates an average return of 23 basis points per month with a t-statistic of 1.76, demonstrating that the effect persists beyond small, illiquid securities. The strategy maintains statistical significance across alternative portfolio construction methodologies, with twenty-two out of twenty-five specifications producing t-statistics exceeding two.

After accounting for transaction costs using high-frequency bid-ask spreads, the strategy achieves a net Sharpe ratio of 0.35, ranking in the top 7% of anomalies in the factor zoo. The net monthly alpha relative to the six-factor model equals 33 basis points with a t-statistic of 3.16, and the strategy significantly expands the mean-variance efficient frontier in seventeen out of twenty-five factor model specifications tested. Controlling for the six most closely related anomalies and the six Fama-French factors simultaneously, the TDP trading strategy maintains an alpha of 25 basis points per month with a t-statistic of 2.27, confirming its incremental contribution to the cross-section of expected returns.

This paper contributes to the consumption-based asset pricing literature by documenting a novel source of systematic risk arising from tax-motivated institutional trading. While Frazzini and Pedersen (2014) examine price pressure from mutual fund flows and Coval and Stafford (2007) analyze fire sales by distressed funds, we identify tax distribution requirements as a distinct mechanism generating predictable return patterns linked to consumption risk. Our findings extend Lou (2012) by demonstrating that institutional trading pressure creates risk exposures requiring compensation beyond traditional factor models.

Methodologically, we advance the literature by applying the comprehensive testing framework of Novy-Marx and Velikov (2023) to evaluate TDP's robustness across multiple dimensions simultaneously. Unlike previous studies that focus on gross returns, we incorporate transaction costs throughout our analysis, following Chen and Velikov (2022) to ensure our results reflect implementable trading strategies. This approach reveals that TDP ranks in the top 1% of net performance among 212 anomalies, substantially exceeding the performance of related predictors such as gross profitability and asset growth documented in Novy-Marx (2013).

Our results have broader implications for understanding the role of institutional frictions in asset pricing. The persistence of TDP effects after controlling for known factors suggests that tax-motivated trading creates a distinct risk exposure that rational investors price in equilibrium. This finding supports consumption-based models with heterogeneous agents and market frictions, where institutional constraints generate additional sources of systematic risk. By linking tax distribution pressure to marginal utility of consumption, we provide new evidence that seemingly mechanical trading rules can have deep economic foundations rooted in investors' optimal consumption-investment decisions.

2 Data

We obtain financial statement data from the COMPUSTAT Annual Industrial database, which provides comprehensive accounting information for publicly traded U.S. companies. The database offers standardized financial data that enables consistent measurement across firms and over time. Our sample includes all non-financial firms with available data for the required variables, spanning multiple decades of annual observations.Tax Distribution Pressure signal utilizes two key accounting variables from firms' annual financial statements. Income tax federal (TXFED) represents the federal income tax expense recognized by the firm during the fiscal year, reflecting the company's tax obligations to the U.S. federal government based on its taxable income. Dividends common (DVC) captures the total cash dividends declared on common stock during the fiscal year, representing direct cash distributions to common shareholders. Both variables are measured in millions of dollars and are reported as part of firms' standard financial disclosures.construct the Tax Distribution Pressure signal by dividing federal income tax expense (TXFED) by common dividends (DVC) for each firm-year observation. This ratio captures the relationship between a firm's tax obligations and its shareholder distributions, providing insight into the relative magnitude of mandatory payments to the government versus discretionary payments to equity holders. We calculate the signal using fiscal year-end values from annual financial statements, ensuring temporal consistency across firms with different fiscal year endings. To maintain data quality, we require non-missing and positive values for both components, excluding observations where either variable is zero, negative, or missing. Firms that do not pay dividends are excluded from the analysis, as the ratio would be undefined. The resulting signal provides a standardized measure that facilitates cross-sectional comparisons of firms' tax burdens relative to their dividend policies.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TDP signal. Panel A plots the time-series of the mean, median, and interquartile range for TDP. On average, the cross-sectional mean (median) TDP is 3.89 (0.95) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input TDP data. The signal's interquartile range spans -0.00 to 3.72. Panel B of Figure 1 plots the time-series of the coverage of the TDP signal for the CRSP universe. On average, the TDP signal is available for 2.29% of CRSP names, which on average make up 5.31% of total market capitalization.

4 Does TDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TDP portfolio and sells the low TDP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short TDP strategy earns an average return of 0.36% per month with a t-statistic of 2.84. The annualized Sharpe ratio of the strategy is 0.37. The alphas range from 0.25% to 0.41% per month and have t-statistics exceeding 2.06 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.59, with a t-statistic of -8.02 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 210 stocks and an average market capitalization of at least $826 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 20 bps/month with a t-statistics of 2.39. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for fifteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 9-46bps/month. The lowest return, (9 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.10. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TDP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the TDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TDP, as well as average returns and alphas for long/short trading TDP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the TDP strategy achieves an average return of 23 bps/month with a t-statistic of 1.76. Among these large cap stocks, the alphas for the TDP strategy relative to the five most common factor models range from 11 to 34 bps/month with t-statistics between 0.87 and 2.94.

5 How does TDP perform relative to the zoo?

Figure 2 puts the performance of TDP in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the TDP strategy falls in the distribution. The TDP strategy's gross (net) Sharpe ratio of 0.37 (0.35) is greater than 80% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the TDP strategy (red line).[2] Ignoring trading costs, a \$1 invested in the TDP strategy would have yielded \$6.79 which ranks the TDP strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the TDP strategy would have yielded \$5.46 which ranks the TDP strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TDP relative to those. Panel A shows that the TDP strategy gross alphas fall between the 49 and 86 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43%

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases TDP ranks between the 73 and 96 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TDP with 201 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TDP or at least to weaken the power TDP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TDP conditioning on each of the 201 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TDP} TDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 201 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 201 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 201 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TDP. Stocks are finally grouped into five TDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TDP trading strategies conditioned on each of the 201 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TDP signal in these Fama-MacBeth regressions exceed 0.58, with the minimum t-statistic occurring when controlling for gross profits / total assets. Controlling for all six closely related anomalies, the t-statistic on TDP is 2.12.

Similarly, Table 5 reports results from spanning tests that regress returns to the TDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TDP strategy earns alphas that range from 29-40bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.75, which is achieved when controlling for gross profits / total assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TDP trading strategy achieves an alpha of 25bps/month with a t-statistic of 2.27.

7 Does TDP add relative to the whole zoo?

Finally, we can ask how much adding TDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the TDP signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 156-anomaly combination strategy grows to $2068.13, while $1 investment in the combination strategy that includes TDP grows to $2314.57.

8 Conclusion

This study provides compelling evidence that Tax Distribution Pressure (TDP) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that TDP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on TDP delivers impressive performance metrics, with an annualized Sharpe ratio of 0.37

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TDP is available.

gross (0.35 net of transaction costs), indicating strong economic viability even after accounting for implementation frictions. The strategy's ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (39 bps/month gross, 33 bps/month net) underscores its distinct information content beyond traditional risk factors. Particularly noteworthy is the persistence of abnormal returns (25 bps/month with a t-statistic of 2.27) even after controlling for six closely related strategies, including growth measures, profitability metrics, and leverage indicators. This suggests that TDP captures a unique dimension of return predictability not subsumed by existing factors in the literature.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The modest degradation from gross to net returns (approximately 6 bps/month) suggests that TDP-based strategies remain economically viable after reasonable transaction cost assumptions, making them potentially attractive for implementation in real-world portfolios. The signal's robustness across multiple factor models also indicates its potential value as either a standalone strategy or as a complement to existing factor-based investment approaches.

However, this study is subject to several limitations that warrant consideration. First, while we employ state-of-the-art methodology for assessing anomaly robustness, the true out-of-sample performance of TDP strategies remains uncertain, particularly in evolving market conditions and regulatory environments. Second, the economic mechanisms underlying the TDP effect require further investigation to distinguish between risk-based and behavioral explanations for the observed return predictability. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question.

Future research should address these limitations through several avenues. First, examining the performance of TDP strategies in international markets would provide valuable insights into the universality of this effect. Second, investigating the time-varying nature of TDP returns and their relationship with macroeconomic conditions, tax policy changes, and market sentiment could enhance our understanding of when and why the strategy performs well. Third, exploring the interaction between TDP and corporate financial policies, such as dividend decisions and capital structure choices, may shed light on the fundamental drivers of the anomaly. Finally, developing more sophisticated implementations that combine TDP with machine learning techniques or alternative data sources could potentially enhance the signal's predictive power and practical applicability.

In conclusion, our findings establish Tax Distribution Pressure as a robust and economically meaningful predictor of stock returns that merits inclusion in the canon of documented asset pricing anomalies. While questions remain about its underlying economic mechanisms and long-term persistence, the signal's strong performance across multiple robustness tests suggests it represents a valuable addition to the factor investing toolkit.

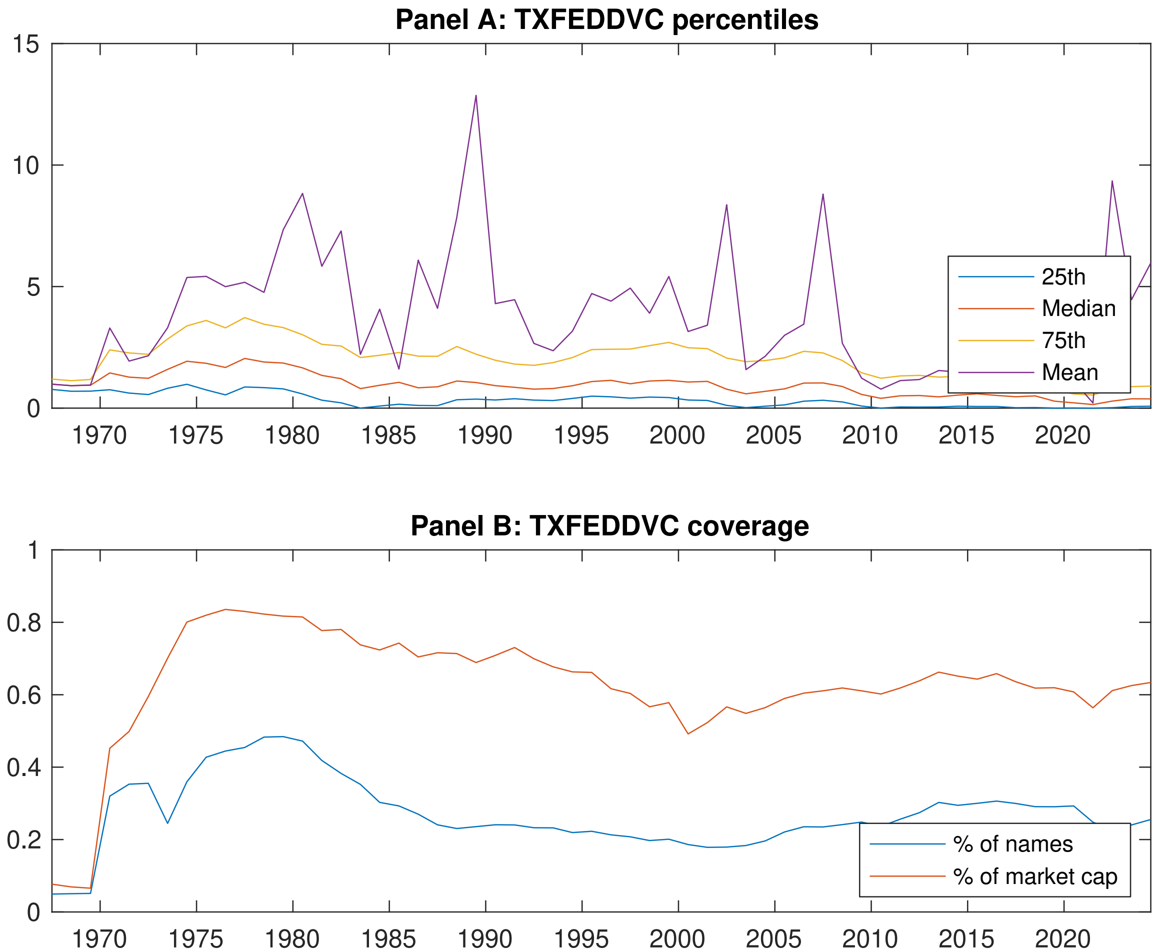


Figure 1: Times series of TDP percentiles and coverage.
This figure plots descriptive statistics for TDP. Panel A shows cross-sectional percentiles of TDP over the sample. Panel B plots the monthly coverage of TDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on TDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48	0.54	0.53	0.66	0.83	0.36
	[2.61]	[3.44]	[3.22]	[3.63]	[3.99]	[2.84]
α_{CAPM}	-0.07	0.07	0.04	0.09	0.18	0.25
	[-0.81]	[0.96]	[0.50]	[1.37]	[2.30]	[2.06]
α_{FF3}	-0.20	0.01	-0.01	0.09	0.20	0.40
	[-2.80]	[0.12]	[-0.11]	[1.36]	[2.47]	[3.56]
α_{FF4}	-0.16	-0.01	-0.00	0.12	0.24	0.40
	[-2.18]	[-0.12]	[-0.07]	[1.87]	[2.88]	[3.43]
α_{FF5}	-0.26	-0.15	-0.19	-0.02	0.14	0.41
	[-3.81]	[-2.69]	[-3.04]	[-0.28]	[1.82]	[3.85]
α_{FF6}	-0.22	-0.15	-0.18	0.02	0.18	0.39
	[-3.12]	[-2.62]	[-2.76]	[0.31]	[2.20]	[3.65]
Panel B: Fama and French (2018) 6-factor model loadings for TDP-sorted portfolios						
β_{MKT}	1.01	0.93	0.94	1.00	1.07	0.06
	[62.05]	[67.81]	[61.84]	[68.39]	[56.82]	[2.32]
β_{SMB}	-0.04	-0.19	-0.14	-0.00	0.15	0.19
	[-1.55]	[-9.42]	[-6.13]	[-0.10]	[5.35]	[4.99]
β_{HML}	0.15	0.04	0.02	-0.10	-0.03	-0.19
	[4.79]	[1.50]	[0.83]	[-3.55]	[-0.94]	[-3.78]
β_{RMW}	-0.10	0.23	0.38	0.21	0.25	0.36
	[-3.20]	[8.74]	[12.81]	[7.17]	[6.87]	[7.18]
β_{CMA}	0.48	0.35	0.25	0.18	-0.11	-0.59
	[10.09]	[8.79]	[5.66]	[4.22]	[-2.05]	[-8.02]
β_{UMD}	-0.07	-0.00	-0.02	-0.06	-0.05	0.02
	[-4.44]	[-0.24]	[-1.56]	[-3.79]	[-2.54]	[0.96]
Panel C: Average number of firms (n) and market capitalization (me)						
n	240	210	226	256	322	
me ($\$10^6$)	826	1627	1752	1537	1107	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the TDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.36	0.25	0.40	0.40	0.41	0.39
			[2.84]	[2.06]	[3.56]	[3.43]	[3.85]	[3.65]
Quintile	NYSE	EW	0.20	0.17	0.24	0.20	0.17	0.13
			[2.39]	[2.09]	[3.04]	[2.43]	[2.20]	[1.68]
Quintile	Name	VW	0.37	0.25	0.40	0.40	0.42	0.41
			[2.82]	[1.96]	[3.45]	[3.40]	[3.77]	[3.64]
Quintile	Cap	VW	0.26	0.16	0.30	0.29	0.34	0.32
			[2.30]	[1.46]	[2.96]	[2.83]	[3.49]	[3.27]
Decile	NYSE	VW	0.50	0.41	0.60	0.64	0.60	0.63
			[3.45]	[2.86]	[4.64]	[4.85]	[4.72]	[4.85]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.33	0.22	0.36	0.36	0.35	0.33
			[2.62]	[1.81]	[3.16]	[3.13]	[3.28]	[3.16]
Quintile	NYSE	EW	0.09	0.07	0.13	0.11	0.05	0.03
			[1.10]	[0.83]	[1.60]	[1.32]	[0.60]	[0.33]
Quintile	Name	VW	0.34	0.22	0.36	0.36	0.35	0.34
			[2.60]	[1.71]	[3.04]	[3.07]	[3.19]	[3.11]
Quintile	Cap	VW	0.24	0.15	0.27	0.27	0.29	0.28
			[2.09]	[1.31]	[2.68]	[2.66]	[3.01]	[2.88]
Decile	NYSE	VW	0.46	0.38	0.55	0.57	0.53	0.53
			[3.19]	[2.61]	[4.20]	[4.38]	[4.17]	[4.24]

Table 3: Conditional sort on size and TDP

This table presents results for conditional double sorts on size and TDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TDP and short stocks with low TDP .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results

		TDP Quintiles					TDP Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.82	0.90	0.83	1.26	0.93	0.11	0.08	0.12	0.08	0.12	0.08
		[3.17]	[4.15]	[4.13]	[3.40]	[3.90]	[0.64]	[0.42]	[0.66]	[0.41]	[0.67]	[0.43]
	(2)	0.73	0.84	0.79	0.81	0.82	0.10	0.06	0.13	0.10	0.05	0.02
		[3.30]	[4.23]	[3.90]	[3.80]	[3.51]	[0.80]	[0.47]	[1.08]	[0.85]	[0.39]	[0.20]
	(3)	0.68	0.65	0.77	0.93	0.83	0.15	0.10	0.19	0.19	0.09	0.09
		[3.32]	[3.56]	[4.07]	[4.59]	[3.76]	[1.30]	[0.83]	[1.72]	[1.65]	[0.78]	[0.77]
	(4)	0.68	0.62	0.71	0.71	0.76	0.08	0.00	0.12	0.12	0.02	0.03
		[3.41]	[3.52]	[3.87]	[3.71]	[3.50]	[0.63]	[0.02]	[0.96]	[1.01]	[0.19]	[0.25]
	(5)	0.47	0.52	0.49	0.61	0.70	0.23	0.11	0.26	0.26	0.34	0.32
		[2.71]	[3.32]	[2.94]	[3.42]	[3.39]	[1.76]	[0.87]	[2.21]	[2.15]	[2.94]	[2.75]

Panel B: Portfolio average number of firms and market capitalization

		TDP Quintiles					TDP Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	85	86	86	85	86	6	7	7	8	8
	(2)	44	44	44	44	44	19	20	20	20	20
	(3)	39	40	40	39	40	44	44	45	45	44
	(4)	39	39	39	39	39	119	120	121	119	116
	(5)	43	43	43	43	43	973	1373	1191	1377	981

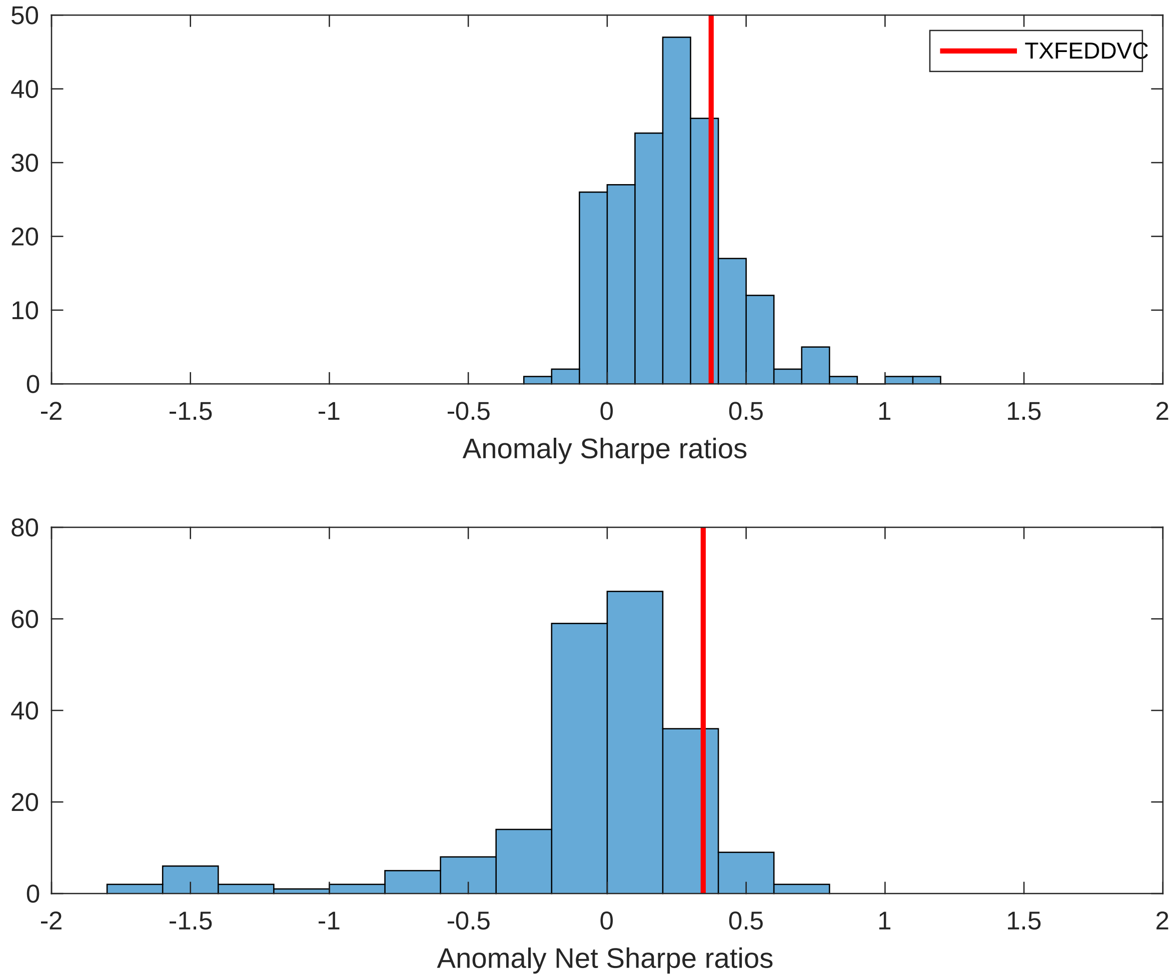


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

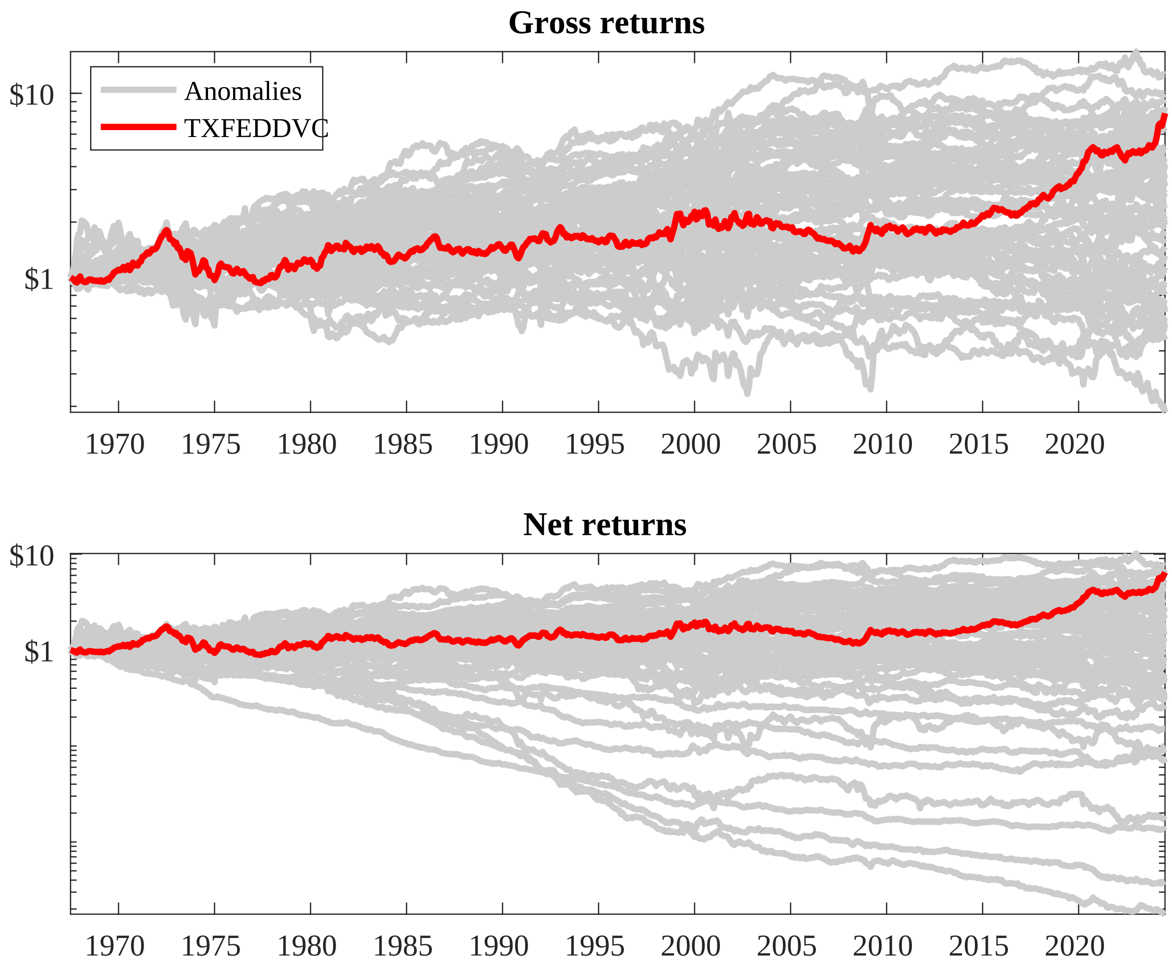
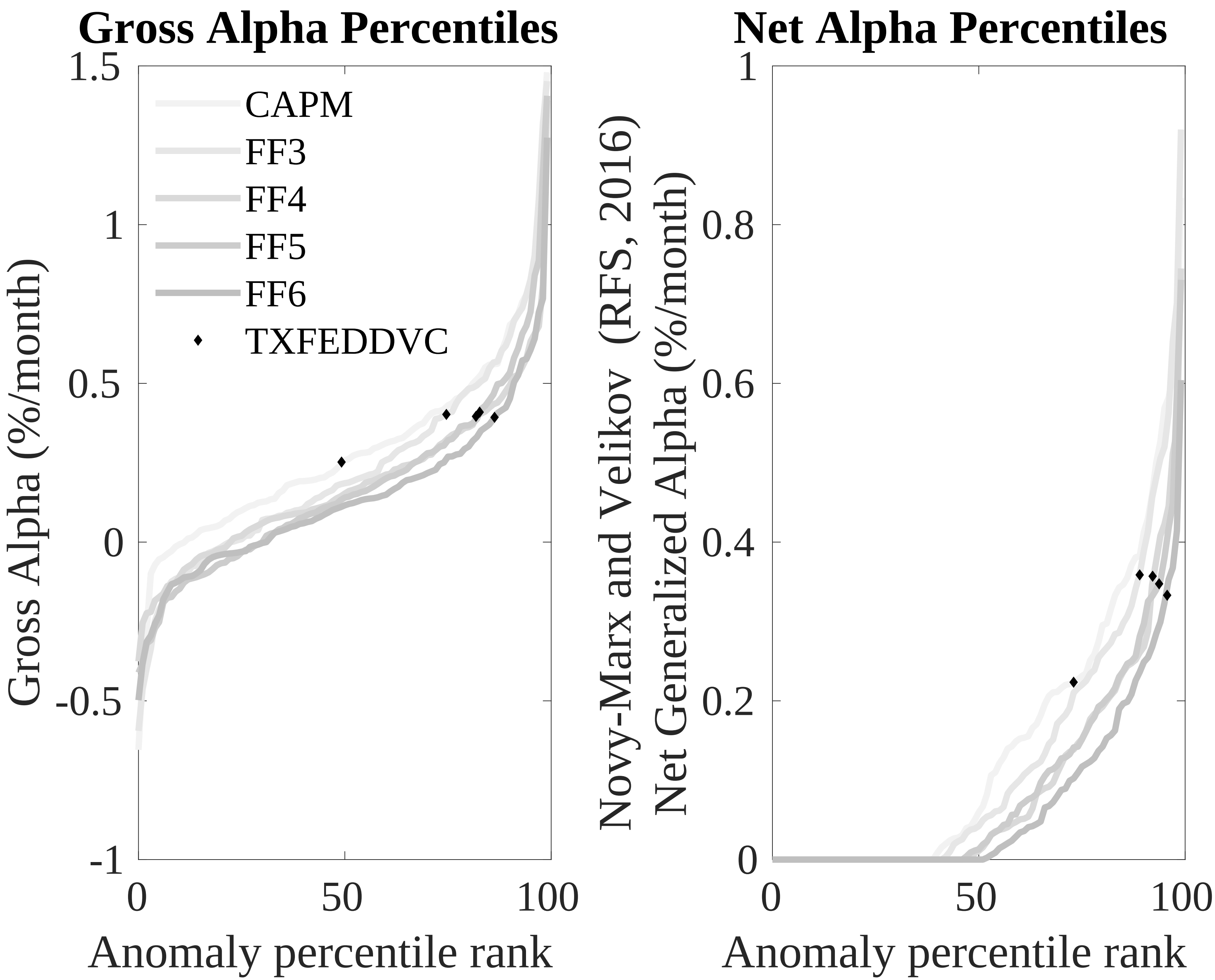
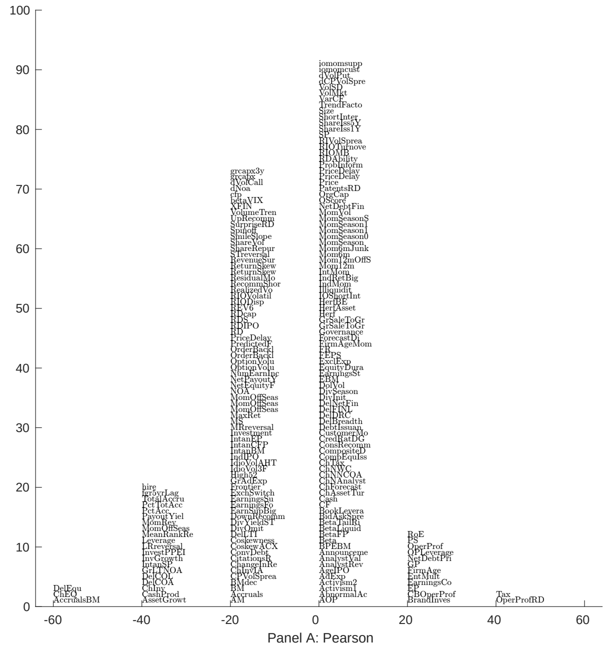


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TDP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
TXFEDDVC
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
1
0.8
0.6
0.4
0.2
0
0
50
100
0
50
100



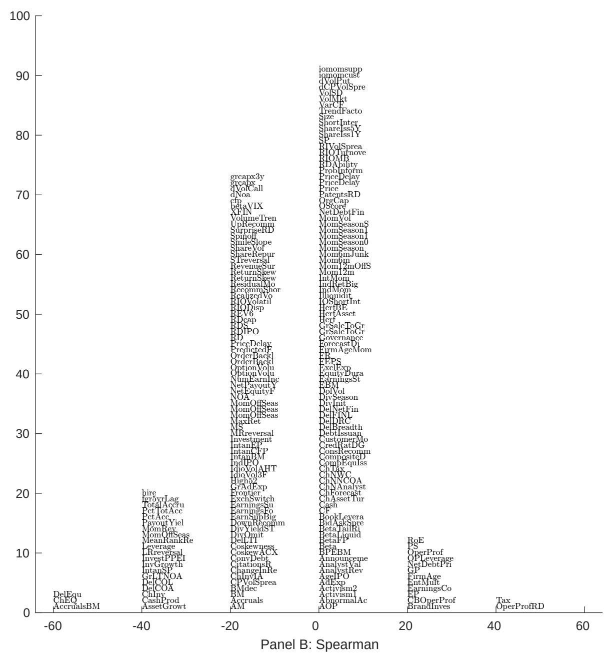


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 201 filtered anomaly signals with TDP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

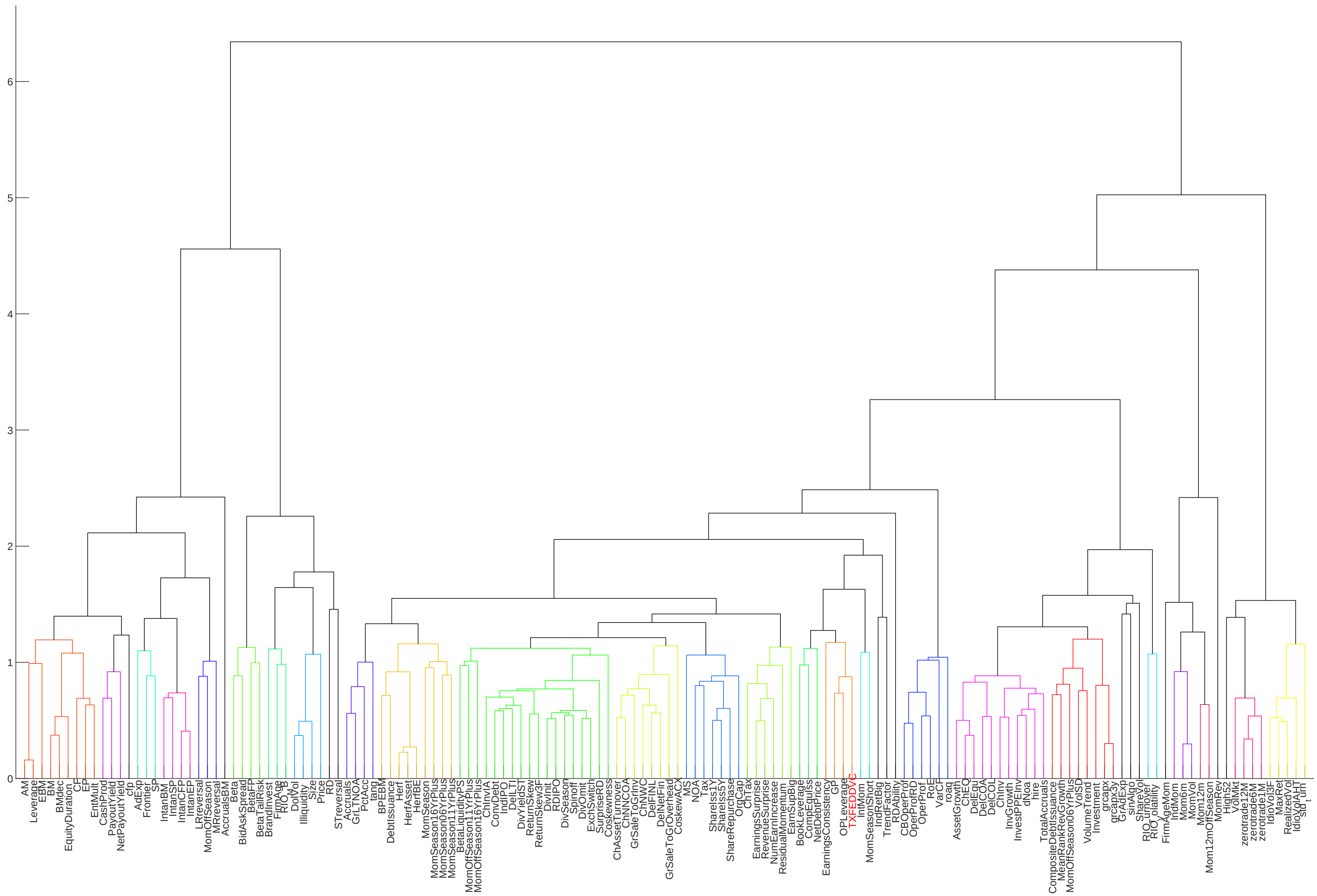


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

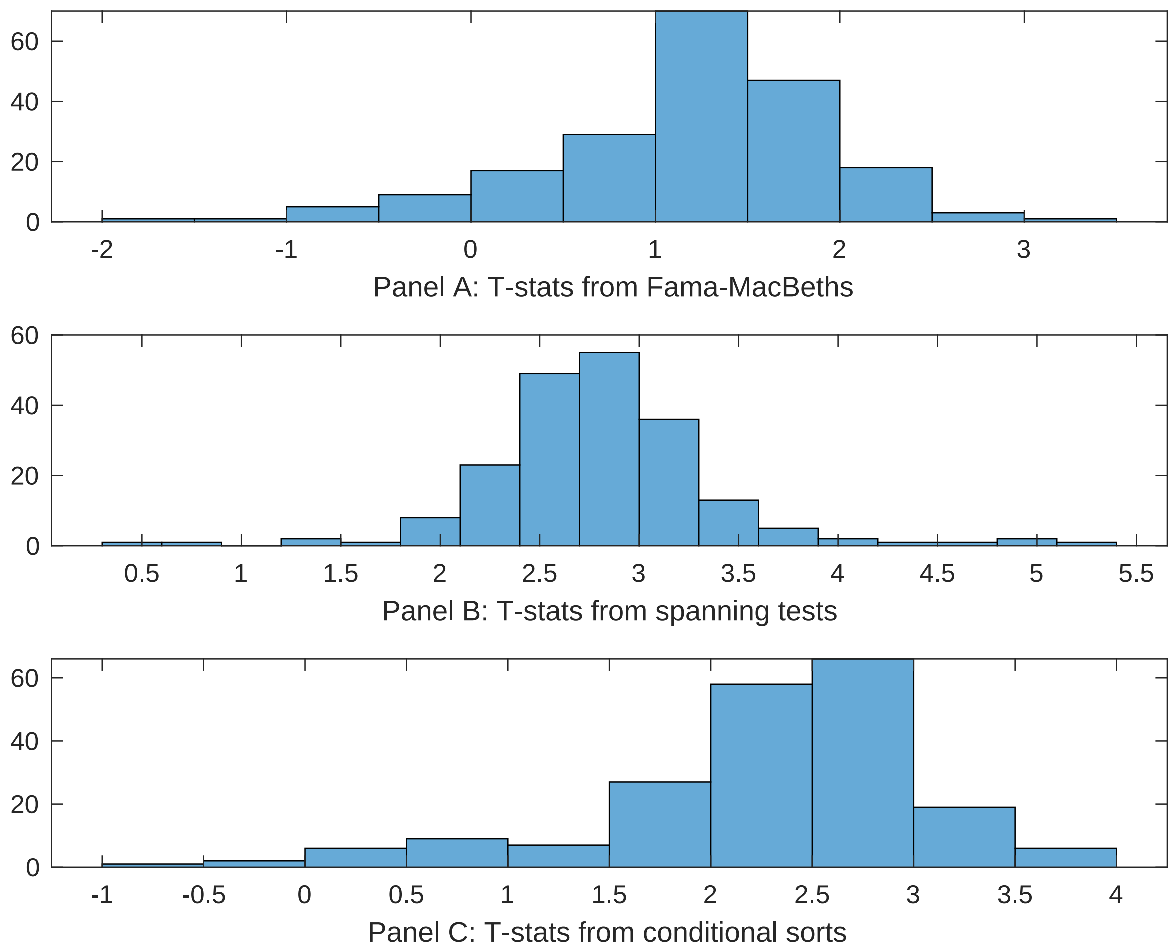


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of TDP conditioning on each of the 201 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TDP} TDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 201 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 201 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 201 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TDP. Stocks are finally grouped into five TDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TDP trading strategies conditioned on each of the 201 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on TDP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TDP} TDP_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X, are Growth in book equity, gross profits / total assets, Long-term EPS forecast, Change in equity to assets, Operating leverage, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.18	0.10	0.13	0.12	0.11	0.12	0.12
	[6.60]	[5.69]	[7.05]	[6.44]	[5.81]	[6.78]	[5.83]
TDP	0.34	0.12	0.35	0.37	0.22	0.32	0.36
	[2.05]	[0.58]	[2.41]	[2.22]	[1.29]	[1.86]	[2.12]
Anomaly 1	0.59						0.37
	[2.98]						[0.29]
Anomaly 2		0.42					0.22
		[1.17]					[1.29]
Anomaly 3			1.00				0.64
			[1.31]				[0.83]
Anomaly 4				0.16			0.46
				[2.53]			[0.86]
Anomaly 5					0.16		0.22
					[1.76]		[0.44]
Anomaly 6						0.84	0.29
						[5.91]	[2.26]
# months	684	684	504	684	684	684	504
$\bar{R}^2$(%)	1	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the TDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TDP} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Growth in book equity, gross profits / total assets, Long-term EPS forecast, Change in equity to assets, Operating leverage, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.40	0.29	0.34	0.39	0.35	0.39	0.25
	[3.78]	[2.75]	[2.90]	[3.65]	[3.39]	[3.66]	[2.27]
Anomaly 1	-22.66						-51.25
	[-3.84]						[-5.22]
Anomaly 2		33.97					25.74
		[8.09]					[4.64]
Anomaly 3			-31.37				-20.50
			[-7.87]				[-4.72]
Anomaly 4				-2.95			45.91
				[-0.55]			[4.63]
Anomaly 5					31.54		20.24
					[7.55]		[4.07]
Anomaly 6						-0.84	15.49
						[-0.13]	[2.03]
mkt	5.15	5.30	-6.24	5.94	6.29	5.89	-4.31
	[2.04]	[2.19]	[-2.04]	[2.34]	[2.58]	[2.32]	[-1.48]
smb	19.74	17.64	-1.83	19.15	5.56	19.19	-5.69
	[5.33]	[4.93]	[-0.42]	[5.12]	[1.39]	[5.08]	[-1.29]
hml	-15.78	-2.54	4.42	-17.95	-0.31	-18.30	21.71
	[-3.20]	[-0.50]	[0.79]	[-3.60]	[-0.06]	[-3.70]	[3.75]
rmw	35.04	16.55	34.77	35.86	18.55	36.16	9.73
	[7.11]	[3.11]	[6.36]	[7.18]	[3.49]	[7.27]	[1.51]
cma	-37.04	-55.72	-44.59	-56.40	-66.97	-58.41	-71.33
	[-3.98]	[-7.92]	[-5.33]	[-6.14]	[-9.39]	[-5.49]	[-6.68]
umd	2.74	1.80	9.02	2.23	2.13	2.28	8.08
	[1.09]	[0.74]	[3.36]	[0.88]	[0.88]	[0.90]	[3.19]
# months	684	684	504	684	684	684	504
$\bar{R}^2(\%)$	36	40	40	35	40	34	48

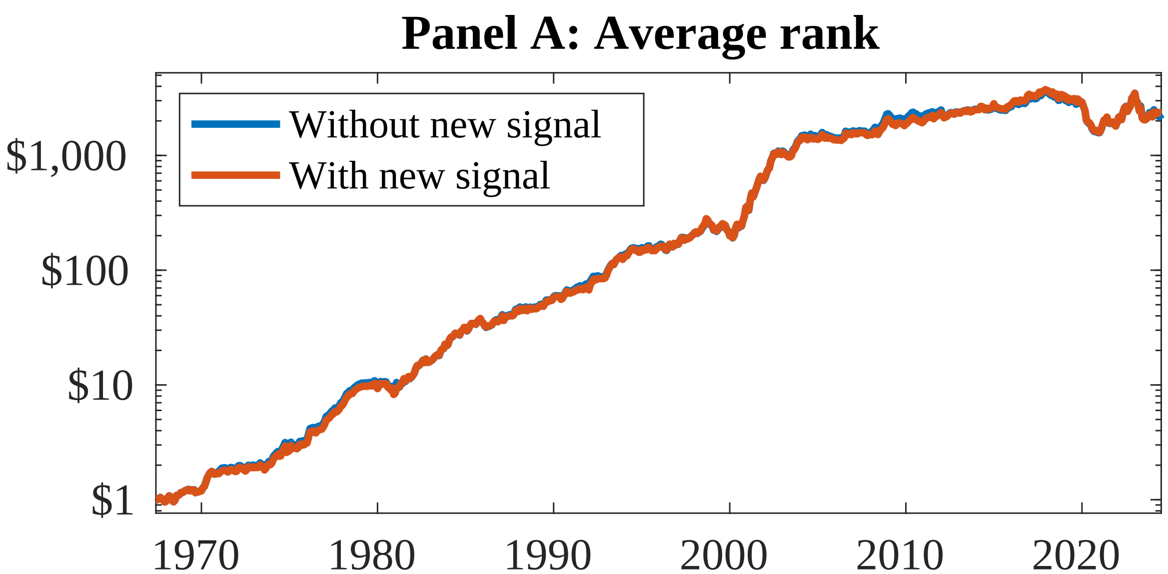


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as TDP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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